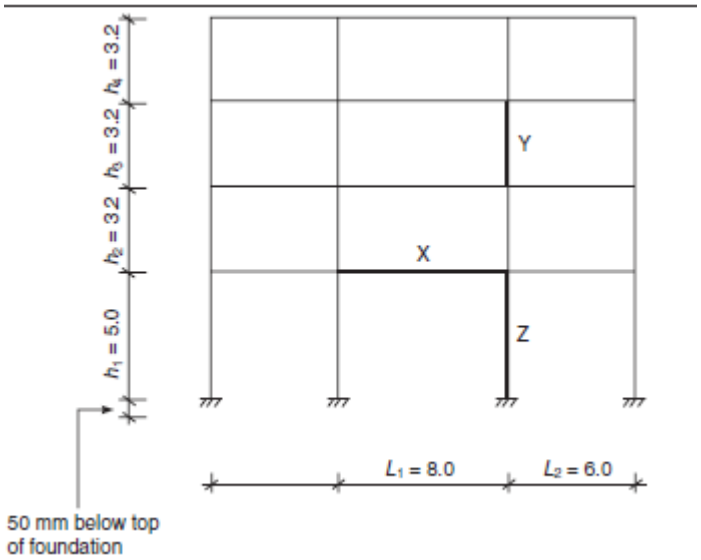
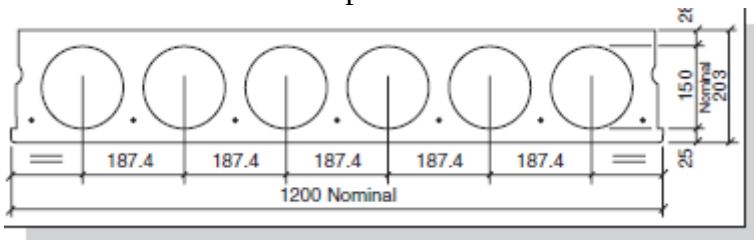
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE**
Course Name	**PRECAST TECHNOLOGY**	**Course Code**	**CIE 4054**
Semester	**VII**	**Max Marks**	**50**
Date	**/ /2024**	**Duration**	**3 Hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No	Question	Marks
1.	Sketch the line of action of major forces between Cast in situ and precast Construction of a simple beam column joint	**4**
2.	Compare between the load Transfer mechanism in braced and unbraced precast frames with suitable sketch	**6**
3.	A symmetrical frame has a central bay of 8m and side bays of 6m. The height of all the floors for upper floors is 3.2m while the ground floor is 5m. Determine, using sub structuring techniques, the bending moments in the beam X and columns Y and Z as depicted in the Figure. The beam–column connections are pinned, and the foundation is rigid. The distance from the edge of the column to the center of the beam end reaction is 100 mm. Characteristic beam loading is gk = 27 kN/m and qk = 21 kN/m. Adopt the partial safety factor of 1.4 for dead load and live load. The cross-sectional dimensions of all the elements are 360mm x 360mm 	**7**

4.	Distinguish between connections and joints.	**3**
5.	Calculate flexural resistance of hollow core floor element with serviceability considerations for the 203mm deep Class 2 prestressed hollow core unit shown in Figure. All the dimensions in the figure are mentioned in mm. The initial prestressing force is 70 per cent of characteristic strength of the 'standard' 7-wire helical strand. Manufacturer's data gives relaxation of 2.5 per cent. Geometric and material data given by the manufacturer are as follows: Area = 135×10^3 mm^2; I = 678×10^6 mm^4; y_t = 99 mm; f_{cu} = 50 N/mm^2; P_c = 30 kN/mm^2; f_{ci} = 35 N/mm^2; P_{ci} = 27 kN/mm^2; f_{pu} = 1750 N/mm^2; P_{ps} = 195 kN/mm^2; A_{ps} = 94.2 mm^2 per strand; cover to 12.5 mm diameter strand = 40 mm. Calculate stresses at the top and bottom extreme fibers of the unit? 	**7**
6.	Explain the need for high strength concrete in prestressed concrete elements.	**3**
7.	Explain the distribution of horizontal loading in a shear wall	**2**
8.	Explain the need for shear walls for skeletal frames greater than 3 floors.	**3**
9.	Write a note on infilled shear wall? Explain the load transfer and sway mechanism in the infill walls	**5**
10.	Illustrate the load transfer at joints with the help of stress contours.	**5**
11.	Explain the structural features of moment resisting connections.	**5**